

An All-Silicon Coherent Optical Receiver Beyond 100 GBd Using a Traveling-Wave SiGe BiCMOS TIA

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Abstract We demonstrate a coherent optical receiver integrating a silicon-photonics front-end with a 55-nm SiGe BiCMOS traveling-wave TIA. The receiver achieves 44-GHz bandwidth and supports single-polarization QPSK up to 140 GBd and 16-QAM up to 112 GBd. ©2026 The Author(s)

Introduction

Coherent optical communication has been widely deployed in long-haul links for decades because it offers high spectral efficiency and enables digital compensation of fiber impairments. For short-reach interconnects, simpler and lower-cost intensity-modulation/direct-detection (IM/DD) links are generally used. However, as data rates continue to scale, coherent technology is expected to become relevant for progressively shorter link spans, including data center interconnects [1], with standards for 800 Gb/s (120 GBd DP-16-QAM) already in place [2].

Compared to IM/DD receivers, coherent receivers require more complex optics to mix the received signal with a local oscillator (LO), as well as broadband and sufficiently linear electronics to support subsequent signal processing. A wide range of integrated coherent receivers has been reported [3–11], but for symbol rates beyond 100 GBd, many coherent receiver implementations rely on InP-based photonic or electronic technologies [12–15] and all-silicon coherent receivers remain scarce: to our knowledge only [16, 17] goes beyond 100 GBd, however no design details were disclosed.

In high-speed receivers, the transimpedance amplifier (TIA) is often implemented using a shunt-feedback (SFB) topology [3–12], because of its favorable trade-off between low noise and high bandwidth (BW). To reach even higher symbol rates, traveling-wave (TW) topologies provide an alternative solution. In this work, a homodyne single-polarization optical coherent receiver front-end is demonstrated up to 140 GBd QPSK and 112 GBd 16-QAM, integrating a TW TIA [18] with a silicon-photonics coherent front-end.

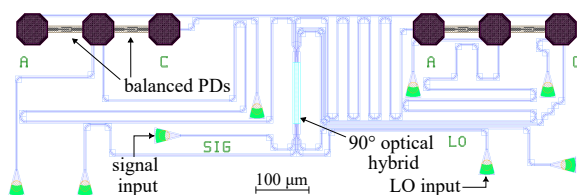


Fig. 1: Layout of the coherent receiver PIC, highlighting key components.

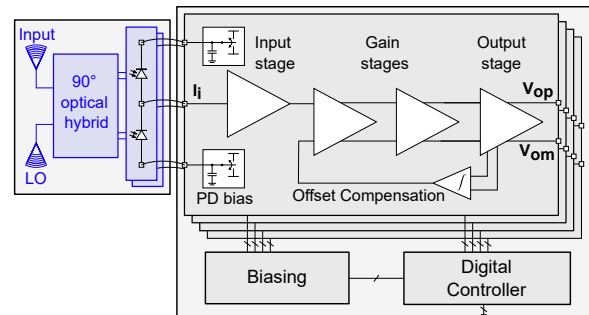


Fig. 2: Block diagram of the 4-channel TIA EIC used with the coherent receiver PIC.

Coherent receiver PIC

The coherent front-end PIC was fabricated in imec's iSiPP200 silicon photonics platform, with its layout shown in Fig. 1. The local oscillator (LO) and signal inputs are coupled into the chip via the indicated grating couplers, after which these optical signals are combined in a 90° optical hybrid, implemented using a 4x4 multimode interferometer (MMI) [19].

The four outputs of this hybrid are routed to two pairs of balanced photodiodes (PDs). Within each pair, the anode of one PD is electrically connected to the cathode of the other, subtracting the common-mode photocurrents on the PIC. The waveguides are routed to avoid crossings and to ensure matched path lengths and bend counts, guaranteeing a minimal difference in delay and loss between the outputs of the optical hybrid and the PDs.

Transimpedance amplifier

The TIA chip employs a traveling-wave (TW) topology to enable high-baudrate operation [18] and includes four TIA channels, each comprising four amplification stages, as illustrated in Fig. 2. With a TIA DC input voltage of around 1.0 V, the balanced PD cathode and anode are biased to 3.5 V and -1.5 V, respectively, to maintain a reverse bias of 2.5 V across each diode, while the TW input stage is able to both sink or source DC photocurrents. These positive and negative bias voltages are supplied externally and each PD bias

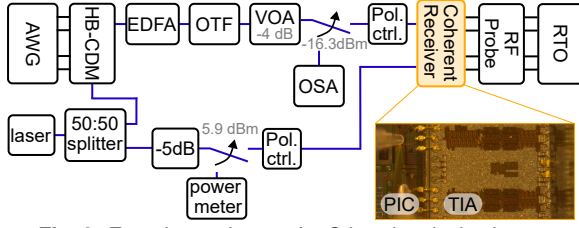


Fig. 3: Experimental setup for C-band optical coherent receiver experiments.

pad can be switched to either of these voltages. The TIA achieves a maximum transimpedance of 76.8 dB Ω (6.9 k Ω) with a corresponding electrical bandwidth beyond 67 GHz [18]. The chip is fabricated using a 55 nm SiGe BiCMOS technology ($f_T = 320$ GHz) and operates from a 2.5 V supply, consuming 500 mW per channel.

To create the full coherent receiver assembly, the PIC was thinned to the same thickness of the TIA chip and diced close to the bondpad edge, after which it was wire bonded to two channels of the TIA. In order to reduce the parasitic inductance, wedge-wedge bonds were used, ensuring a minimal loop height.

Experimental setup

This assembly was then integrated in the measurement setup shown in Fig. 3. The arbitrary waveform generator (AWG) generates the electrical in-phase (I) and quadrature (Q) waveforms based on a PRBS-11 bit stream, mapped to the desired constellation and pulse-shaped by a root-raised cosine (RRC) filter with a roll-off factor of 0.35. These differential I and Q signals are then amplified and modulated onto a C-band optical carrier using an 80 GHz high-bandwidth coherent driver/modulator (HB-CDM)[20]. Although this transmitter supports dual-polarization operation, only a single polarization was used in this experiment. An automatic bias controller (ABC) was employed to bias the I and Q child Mach-Zehnder modulators (MZMs) at their minimum transmission point, while the parent MZM is biased at its quadrature bias point.

The modulated optical signal was amplified by an erbium-doped fiber amplifier (EDFA) and subsequently filtered by an optical tunable filter (OTF) with a bandwidth of approximately 1.5 nm (190 GHz) to suppress out-of-band noise. A variable optical attenuator (VOA) was used to adjust the optical power incident on the receiver. An optical switch enabled the signal to be directed either to an optical spectrum analyzer (OSA) for spectral analysis or to the coherent receiver PIC via a polarization controller. Throughout the experiment, the VOA attenuation was set to 4 dB, for which its optical output power was measured at -16.3 dBm. The LO signal was derived from the same <0.1 kHz linewidth laser (Koheras Basik), us-

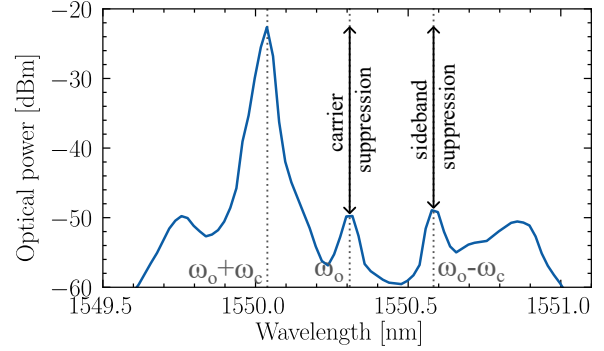


Fig. 4: OSA measurement of the optical spectrum for $f_c = 35$ GHz when the transmitter skew is properly compensated.

ing a 50:50 optical power splitter. A 5 dB optical attenuator was used to reduce the LO power to 5.9 dBm before coupling it into the PIC through a second polarization controller.

The two differential outputs of the TIA are probed simultaneously using a custom-designed RF probe with two ground-signal-signal-ground (GSSG) pairs, connected to the four inputs of a Keysight UXR1104A real-time oscilloscope (RTO). For the measurement results presented in this work, only the two middle signals (corresponding to a single-ended output from each differential pair) were utilized. This choice was made because the outermost RTO ports were connected via longer coaxial cables, which introduced additional attenuation, degrading the measurement quality.

Skew compensation

The drive strength of the two AWG channels was individually adjusted while monitoring the output power of the transmitter, ensuring equal optical power levels for the I and Q components. Subsequently, a $\cos(\omega_c t)$ and $\sin(\omega_c t)$ waveform were applied at the I and Q channel of the AWG. This configuration enables compensation of the skew between the I and Q waveforms by introducing a delay on one of the AWG outputs while observing the resulting optical spectrum on the OSA. When the skew is properly compensated, a single-sideband (SSB) response is observed, as predicted by the following identity, where ω_o and Δt denote the angular frequency of the optical carrier and the skew to be compensated, respectively:

$$\begin{aligned}
 & e^{j\omega_o t} \cdot \cos\left(\omega_c\left(t + \frac{\Delta t}{2}\right)\right) + j e^{j\omega_o t} \cdot \sin\left(\omega_c\left(t - \frac{\Delta t}{2}\right)\right) \\
 &= e^{j(\omega_o + \omega_c)t} \cdot \underbrace{\cos\left(\omega_c \frac{\Delta t}{2}\right)}_{\rightarrow 1 \text{ if } \Delta t \rightarrow 0} - j e^{j(\omega_o - \omega_c)t} \cdot \underbrace{\sin\left(\omega_c \frac{\Delta t}{2}\right)}_{\rightarrow 0 \text{ if } \Delta t \rightarrow 0}
 \end{aligned}$$

Fig. 4 presents the measured optical spectrum following this calibration procedure, with $f_c = 35$ GHz. The minimum transmission point biasing gives rise to carrier suppression (around 26 dB), while the sideband suppression (around 25 dB)

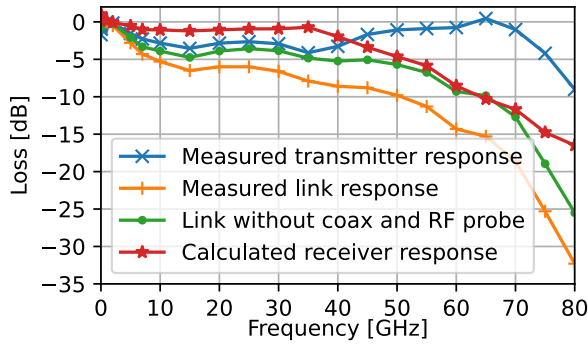


Fig. 5: Frequency response of (sections of) the coherent link, normalized at 1 GHz.

is due to the skew calibration. To ensure the skew compensation is correct across the entire frequency range, this procedure may need to be repeated at multiple frequencies f_c .

With the transmitter skew properly compensated, the receiver-side skew can be corrected at the RTO. This is achieved by analyzing the phase of the fast-fourier transform (FFT) of each received signal and adjusting the delay settings of the RTO channels.

Link frequency response

By sweeping the modulation frequency f_c in this SSB configuration, the frequency response of the coherent optical link and its constituent components can be characterized, as shown in Fig. 5. The transmitter response is obtained by measuring the optical output power directly at the transmitter, while the overall system response is derived from the FFT of the received signal, captured by the RTO. The known losses of the RF probe and the 10 cm coaxial cable to the RTO can be de-embedded from the measured response, and by further de-embedding the transmitter response, the frequency response of the receiver can be extracted. This analysis reveals that the receiver exhibits a 3-dB bandwidth of approximately 44 GHz. This is lower than the 64 GHz bandwidth achieved when using this TIA as an IM/DD receiver [18], which can be attributed to the use of balanced photodiodes in the coherent receiver, which doubles the impact of the photodiode parasitics.

Measurement results

To extract the data from the waveforms captured by the RTO, offline DSP is performed. The received signals are first passed through an RRC filter with a roll-off factor of 0.35, matched to the transmit-side pulse shaping. Subsequently, the Gram-Schmidt orthogonalization procedure is applied to correct for any I/Q imbalance in amplitude or phase. Following this, the waveforms are re-sampled to two samples per symbol, and clock recovery is performed, after which a 30-tap equalizer is applied to the data. Since the same laser is used for both the transmitter and the LO in this ho-

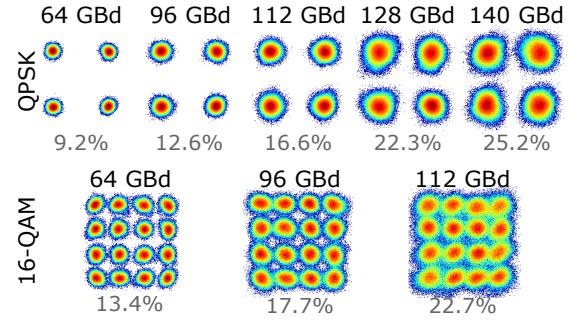


Fig. 6: Received QPSK and 16-QAM constellation diagrams, with logarithmic color mapping and EVM values annotated.

modity setup, the received signal does not have any frequency offset, eliminating the need for frequency recovery. After a final phase recovery step, the received data can be decoded.

The received quadrature phase shift keying (QPSK) constellations are shown at the top of Fig. 6. To quantitatively assess the performance of these results, the error vector magnitude (EVM) is calculated as the RMS average of the distances between the received symbols and their ideal constellation points, normalized to the RMS average amplitude of the constellation points. These values are annotated below the constellation diagrams, showing a clear trend of increasing EVM with higher symbol rates. For these QPSK results, performance is also evaluated by counting the bit errors. Up to and including 112 GBd (224 Gb/s gross), no bit errors were observed in the entire captured data stream of 1.92 million symbols. At 128 GBd (256 Gb/s gross), a bit-error rate (BER) of 1.2×10^{-3} is measured, which remains below the threshold for 6.7% overhead hard-decision forward error correction (HD-FEC). At 140 GBd (280 Gb/s), the BER increases to 9.5×10^{-3} , within the correction capability of 25% overhead soft-decision FEC (SD-FEC).

Additional measurement results with 16-QAM modulation are displayed on the bottom of Fig. 6. 16-QAM constellation diagrams for 64 GBd, 96 GBd and 112 GBd, corresponding to gross bit rates of 256 Gb/s, 384 Gb/s and 448 Gb/s, respectively, are shown with their EVM values annotated.

Conclusions

A coherent optical receiver combining a silicon photonic coherent front-end with a SiGe BiCMOS traveling-wave TIA was demonstrated. The receiver exhibits a 3-dB bandwidth of 44 GHz and supports single-polarization QPSK reception up to 140 GBd (280 Gbit/s gross) and 16-QAM up to 112 GBd (448 Gbit/s gross). These results highlight the potential of traveling-wave TIAs for coherent receivers beyond 100 GBd and show that silicon photonics combined with SiGe BiCMOS electronics is a promising platform for future high-speed integrated coherent links.

Acknowledgements

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References

- [1] X. Zhou, "IM-DD vs. Coherent in Datacenters: A Revisit in 2025", in *2025 Optical Fiber Communications Conference and Exhibition (OFC)*, Mar. 2025, pp. 1–28.
- [2] K. Kota, D. R. Stauffer, and K. Gass. "800ZR Coherent Implementation Agreement". [Online]. Available: <https://www.oiforum.com/wp-content/uploads/OIF-800ZR-01.0.pdf>.
- [3] M. G. Ahmed, T. N. Huynh, C. Williams, Y. Wang, P. K. Hanumolu, and A. Rylyakov, "34-GBd Linear Transimpedance Amplifier for 200-Gb/s DP-16-QAM Optical Coherent Receivers", *IEEE Journal of Solid-State Circuits*, vol. 54, no. 3, pp. 834–844, Mar. 2019. DOI: 10.1109/JSSC.2018.2882265.
- [4] A. Awany et al., "23.5 A dual 64Gbaud 10kΩ 5% THD linear differential transimpedance amplifier with automatic gain control in 0.13μm BiCMOS technology for optical fiber coherent receivers", in *2016 IEEE International Solid-State Circuits Conference (ISSCC)*, Jan. 2016, pp. 406–407. DOI: 10.1109/ISSCC.2016.7418079.
- [5] L. A. Valenzuela, Y. Xia, A. Maharry, H. Andrade, C. L. Schow, and J. F. Buckwalter, "A 50-GBaud QPSK Optical Receiver with a Phase/Frequency Detector for Energy-Efficient Intra-data Center Interconnects", *IEEE Open Journal of the Solid-State Circuits Society*, pp. 50–60, 2022. DOI: 10.1109/OJSSCS.2022.3150291.
- [6] A. Wissing et al., "A 100 Gbps, sub-pJ/bit Transimpedance Amplifier in 90-nm SiGe in a Reconfigurable IMDD/Coherent Optical Receiver", in *2025 Optical Fiber Communications Conference and Exhibition (OFC)*, Mar. 2025, pp. 1–3. DOI: 10.1364/OFC.2025.M2H.3.
- [7] J. Liu et al., "DSP-Free Optical Phase Locked Loop Receiver for Energy Efficient Coherent Intra-Data Center Communications", in *2024 European Conference on Optical Communication (ECOC)*, Frankfurt, 2024.
- [8] G. Movaghar et al., "A Monolithic O-Band Coherent Optical Receiver for Energy-Efficient Links", *IEEE Journal of Solid-State Circuits*, vol. 59, no. 5, pp. 1409–1420, May 2024. DOI: 10.1109/JSSC.2023.3339494.
- [9] J. Verbist et al., "A 40-GBd QPSK/16-QAM Integrated Silicon Coherent Receiver", *IEEE Photonics Technology Letters*, vol. 28, no. 19, pp. 2070–2073, Oct. 1, 2016. DOI: 10.1109/LPT.2016.2582799.
- [10] P. M. Seiler et al., "Multiband Silicon Photonic ePIC Coherent Receiver for 64 GBd QPSK", *Journal of Lightwave Technology*, vol. 40, no. 10, pp. 3331–3337, May 15, 2022. DOI: 10.1109/JLT.2022.3158423.
- [11] A. H. Ahmed, L. Vera, L. Iotti, R. Shi, S. Shekhar, and A. Rylyakov, "A Dual-Polarization Silicon-Photonic Coherent Receiver Front-End Supporting 528 Gb/s/Wavelength", *IEEE Journal of Solid-State Circuits*, vol. 58, no. 8, pp. 2202–2213, Aug. 2023. DOI: 10.1109/JSSC.2023.3257103.
- [12] S. Kumagai et al., "A Shunt-Feedback TIA with Common-Base Variable Gain Input Stage for 128-GBaud Coherent Communication", in *2023 IEEE BiCMOS and Compound Semiconductor Integrated Circuits and Technology Symposium (BCICTS)*, Oct. 2023, pp. 163–166. DOI: 10.1109/BCICTS54660.2023.10310925.
- [13] Y.-W. Chen et al., "InP CDM and ICR Enabled 128Gbaud/ DP-16QAM-PS and 120Gbaud/DP-QPSK Long-Haul Transmission", *IEEE Photonics Technology Letters*, vol. 34, no. 9, pp. 471–474, May 1, 2022. DOI: 10.1109/LPT.2022.3165484.
- [14] J. Gläsel et al., "128 GBaud Coherent Receiver Engine with Flat Frequency Response", in *2024 Optical Fiber Communications Conference and Exhibition (OFC)*, Mar. 2024, pp. 1–3. DOI: 10.1364/OFC.2024.W1K.7.
- [15] E. Rouvalis et al., "Integrated Coherent Transmit-Receive Optical Sub-Assembly (IC-TROSA) for 140 GBd Applications", in *2024 Optical Fiber Communications Conference and Exhibition (OFC)*, Mar. 2024, pp. 1–3. DOI: 10.1364/OFC.2024.W3A.2.
- [16] X. Lei et al., "C+L 800G Integrated Coherent Receiver Based on Silicon Photonics", in *2024 Asia Communications and Photonics Conference (ACP) and International Conference on Information Photonics and Optical Communications (IPOC)*, Nov. 2024, pp. 1–3. DOI: 10.1109/ACP/IP0C63121.2024.10809792.
- [17] X. Lei et al., "High-Performance Heterogeneously Integrated Coherent Optical Sub-Assembly Enabling 130 Gbaud DP-QPSK Transmission", in *2025 European Conference on Optical Communications (ECOC)*, Sep. 2025, pp. 1–4. DOI: 10.1109/ECOC66593.2025.11263018.
- [18] J. Declercq et al., "A 64-GHz Optical Receiver for 128-GBd Links Using a 55-nm SiGe BiCMOS Traveling-Wave Linear Transimpedance Amplifier", *Journal of Lightwave Technology*, vol. 43, no. 9, pp. 4139–4148, May 2025. DOI: 10.1109/JLT.2025.3533200.
- [19] J. Zhang et al., "Compact Low-Power-Consumption 28-Gbaud QPSK/16-QAM Integrated Silicon Photonic/Electronic Coherent Receiver", *IEEE Photonics Journal*, vol. 8, no. 1, pp. 1–10, Feb. 2016. DOI: 10.1109/JPHOT.2015.2505609.
- [20] K. Kota, D. R. Stauffer, and K. Gass. "OIF-HB-CDM-02.0: Implementation Agreement for the High Bandwidth Coherent Driver Modulator". [Online]. Available: <https://www.oiforum.com/wp-content/uploads/OIF-HB-CDM-02.0.pdf>.